Construction

Time Limit

7min

Assembling Rebar

Watch the video

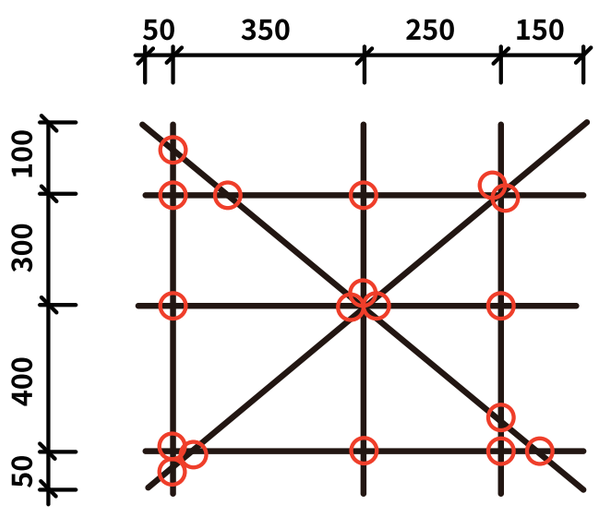
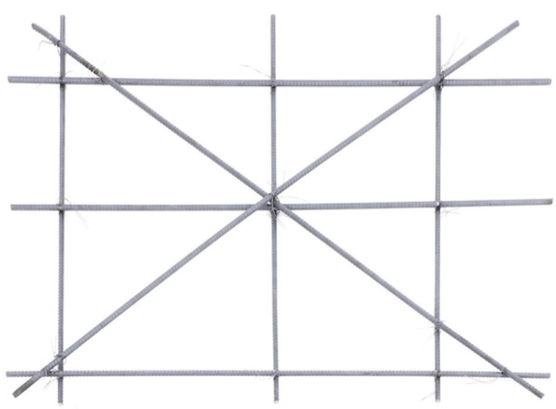
Job Category	Construction	Subject	Assembling Rebar

Ⅰ Requirements

– Perform the rebar fixing within the given time according to the diagram.

Ⅱ Diagram & Requirements

1) Diagram

단위 : mmUnit : mm	Setting Example
	

Job Category	Construction	Subject	Assembling Rebar

2) Requirements

① Each task involving a tying wire or handle must be performed accurately.

② Make sure that the tying location and the number of ties are in accordance with the requirements.

③ Bind the every binding point.

④ Make sure that your work is in accordance with the specification on the diagram.

⑤ After the evaluation, disassemble wour works.

❖ Failure to wear safety equipment, such as gloves, or removing it during work will result in the lowest score.

Job Category	Construction	Subject	Assembling Rebar

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Distinctiron bar	· D10 X 800mm	PCS	3	For 1 person
		· D10 X 1,000mm		3	
		· D10 X 1,200mm		2	
	Wire for fixing	· #20번 길이(350mm)	PCS	40	For 1 person
	Fixing tool	· For iron bar fastening	PCS	1	For 1 person
	Ruler	· 1M or more	PCS	1	For 1 person
	Writing tool	· For indicating the binding position (chalk or white marker)	PCS	1	For 1 person
	Nipper	· For cutting Wire for fixing	PCS	1	For 1 person
	Safety Equipment	· Safety helmet, safety gloves	PCS	1	For 1 person

Time Limit

10min

Pipe Scaffold Install/Dismantle

Watch the video

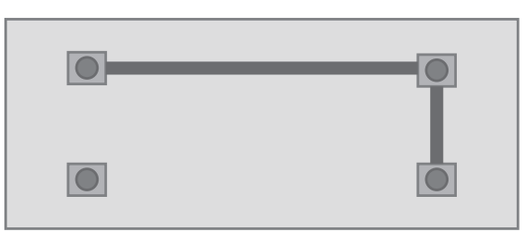
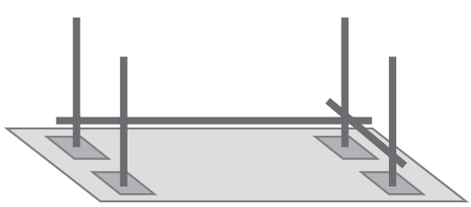
Job Category	Construction	Subject	Pipe Scaffold Install/Dismantle

I Requirements

– Install and dismantle the steel pipe scaffold using tools.

II Diagram & Requirements

1) Diagram

	The shape from above	The shape from the front
Initial State		
	Half of the structure is pre-finished(pre-set), and the other half wil be tested.	

How to work



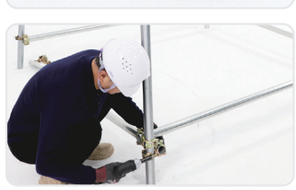


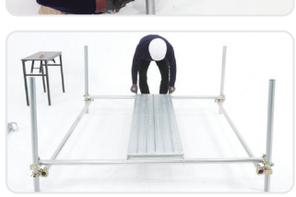












Job Category	Construction	Subject	Pipe Scaffold Install/Dismantle

2) Requirements

① A-3).Attach clamp to the verticaly installed post(A-1), fasten the bolt(A-2), and use an impact wrench to tighten the nut.(A-3)

② Connect the pipe horizontally to the opposite clamp(B-1). Install an additional clamp just above the fastened pipleand connect the pipe in a cross shape(B-2). Be careful not to fasten pipe to the outside of the installed vertical pipe.

③ C-3). Use the leveler to check the horizontal level(C-1). Connect the other 2 pipes in the same way(C-2) and insert the workbench between the posts.

④ After the test, perform the dismantling operation in the pre-test state followed the instructor.

Job Category	Construction	Subject	Pipe Scaffold Install/Dismantle

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Pipe	· Standard	PCS	2	For 1 person
	Clamp	· Standard	PCS	4	For 1 person
	Workbench	· Standard	PCS	1	For 1 person
	Impact Wrench	· Standard	PCS	1	For 1 person
	Horizontal Ruler	· Standard	PCS	1	For 1 person

Time Limit

2min

Using Grinders

Watch the video

Job Category	Construction	Subject	Using grinders

I Requirements

– Understand how to combine / disassemble and use the grinder.

II Diagram & Requirements

1) Diagram



Job Category	Construction	Subject	Using grinders

2) Requirements

① Attach the grinder handle.

② Attach the guard cover to the grinder and use a hex wrench to tighten the screw completely.

③ Insert the flange(만조), disc, and clamping nut in that order.

④ Press the spindle lock on the grinder to lock the rotating shaft.

⑤ Turn on the grinder, perform a usage simulation, then turn off the grinder.

⑥ After the supervisor's confirmation, if instructed, press the spindle lock and disassemble in the reverse order of assembly.

Ⅲ Equipment & Materials

Equipment & Materials	Spec.	Unit	QTY	Remakrs
 Grinder Set	· 4inch Grinder · Handle · Guard Cover · Disc · Clamping Nut · Grinder Spanner · Hex Wrench	SET	1	For 1 person

Time Limit

7min

Pipe Volting

Watch the video

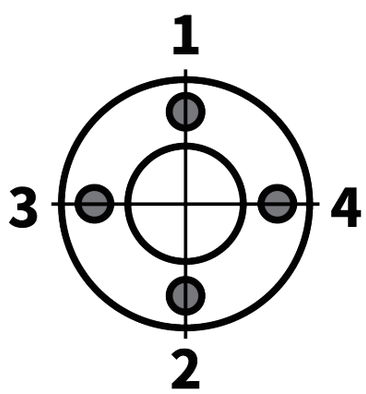
Job Category	Construction	Subject	Pipe Volting

I Requirements

– Use the tool to join the piping flanges.

II Diagram & Requirements

1) Diagram

Assemble Sequence	Example
	
Result	**Result (Side)**
	

Job Category	Construction	Subject	Pipe Volting

2) Requirements

① Assemble the flange using the provided materials and tools, and obtain confirmation from the supervising officer.

<order of assembly>

1. Insert the gasket between the flanges and align it.
2. Lightly tighten the bolts, then check the offset between the two flanges.
3. Using a spanner, tighten the bolts and nuts in a diagonal sequence.
4. During the final tightening, confirm that there is no offset on the flange to complete the task.

② After the supervisor's confirmation, disassemble the assembled flange and gasket within the time limit according toinstructions.

Job Category	Construction	Subject	Pipe Volting

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Flange	· 5K 50A	PCS	2	For 1 person
	Gasket	· 5K 50A	PCS	1	For 1 person
	Hex Bolt	· M12, 45mm	PCS	4	For 1 person
	Hexagon Nut	· M12	PCS	4	For 1 person
	Washer	· M12	PCS	8	For 1 person
	Spanner	· 19mm	PCS	2	For 1 person
	Vernier Calipers	· Digital Vernier Calipers · Replace with a ruler if local supply is not available	PCS	1	For 1 person

Time Limit

2min

Wire Connection

Watch the video

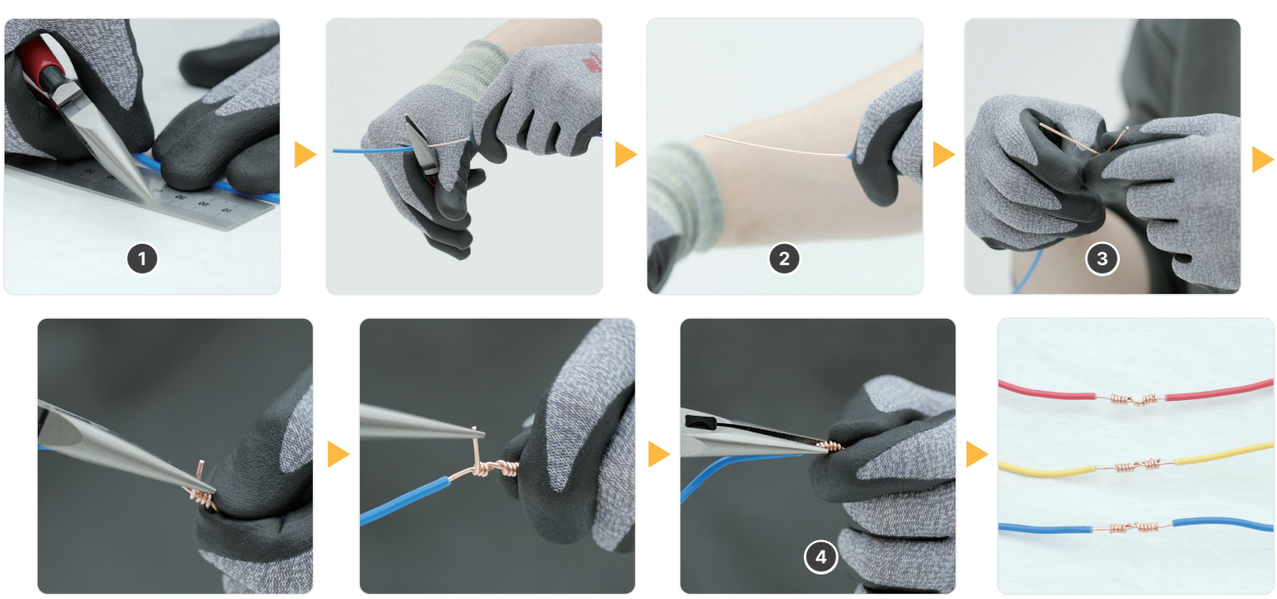
Job Category	Construction	Subject	Wire Connection

I Requirements

– Peeled off the sheath of the wire and connect to wires of the same color.

II Diagram & Requirements

1) Diagram



Job Category	Construction	Subject	Wire Connection

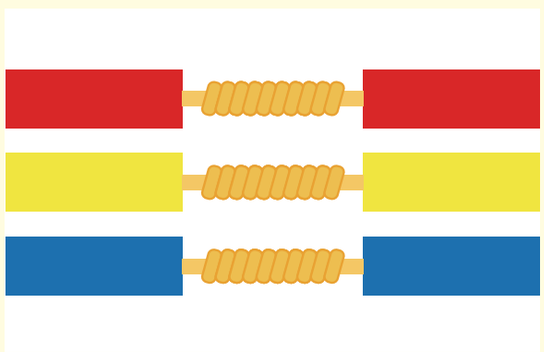
2) Requirements

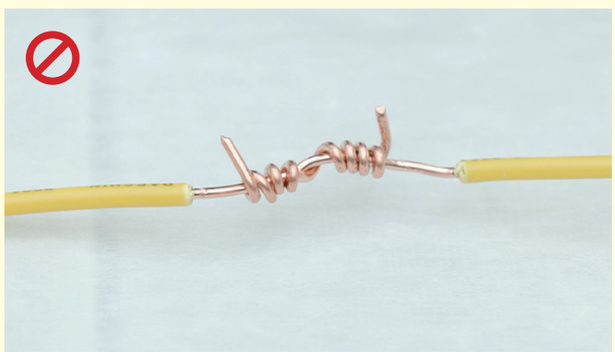
① Using long-nose pliers, peel off the sheath of the given wire about 6 cm.

② Other wires too.

③ Using long-nose pliers, twist the two copper wires together. (Wires of same color must be tied)

Ex)





④ Secure the ends of the twisted copper wires using long-nose pliers.

❖ A total of six wires are provided, consisting of two red, two yellow, and two blue wires.

Job Category	Construction	Subject	Wire Connection

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Long-nose Pliers	· Standard	PCS	1	For 1 person
	HIV HIV wire	· 2.5 SQ, 6 per person (2 per color) · Colors: red, yellow, blue · Min 30 cm or more, 1.5 to 2 mm thick	PCS	6	For 1 person
	Ruler	· 30cm 30cm or more	PCS	1	For 1 person
	Safety Gloves	· Standard	PCS	1	For 1 person